PORTABLE IOT-BASED SLEEP APNEA MONITORING SYSTEM FOR HOME HEALTHCARE

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BONAFIDE CERTIFICATE

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ABSTRACT

Sleep Apnea is a serious sleep disorder characterized by repeated interruptions in breathing during sleep, leading to reduced oxygen levels and increased risk of cardiovascular diseases. Traditional diagnosis methods such as Polysomnography (PSG) are expensive, complex, and not easily accessible for continuous monitoring. To overcome these limitations, this project proposes a Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare. The system is designed using an ESP32 microcontroller integrated with an IR sensor to monitor physiological signals such as heart rate and SpO_2 levels. The collected data is processed in real-time and displayed on an LCD screen for immediate user feedback. The ESP32 utilizes its built-in WIFI module to transmit the data to the Thing Speak cloud platform, enabling continuous remote monitoring and visualization through graphical analysis. Sleep Apnea is a potentially life-threatening sleep disorder characterized by repeated interruptions in breathing during sleep, leading to oxygen desaturation and disturbed sleep patterns. If left undiagnosed, it can result in serious health complications such as cardiovascular diseases, hypertension, and chronic fatigue. Traditional diagnostic methods like Polysomnography (PSG) are expensive, complex, and require clinical supervision, making them unsuitable for continuous and home-based monitoring. To address these challenges, this project proposes a Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare. The system is designed using an ESP32 microcontroller integrated with an IR sensor to continuously monitor physiological parameters such as heart rate and oxygen levels (SpO_2). The sensor data is processed in real-time using a threshold-based algorithm implemented in embedded C through the Arduino IDE. The processed data is displayed locally on an LCD screen for instant feedback and is simultaneously transmitted to the Thing Speak cloud platform using the ESP32's built-in WIFI module. The cloud platform enables real-time visualization of health parameters through graphical representation and allows users to download the data in Excel.

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CHAPTER 1

INTRODUCTION

Sleep Apnea is a common yet serious sleep disorder characterized by repeated interruptions in breathing during sleep. These interruptions reduce oxygen levels in the blood and disturb normal sleep patterns, leading to various health complications such as hypertension, cardiovascular diseases, daytime fatigue, and reduced concentration. Traditional diagnostic methods like Polysomnography (PSG) are widely used in hospitals for accurate detection; however, they are expensive, complex, and require continuous medical supervision. This makes them unsuitable for regular monitoring, especially for patients who require long-term observation in home environments. To overcome these limitations, this project proposes a Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare. The system utilizes an ESP32 microcontroller integrated with an IR sensor to continuously monitor physiological parameters such as heart rate and oxygen levels (SpO_2). The collected data is processed in real-time using a threshold-based algorithm and displayed locally on an LCD screen for immediate user awareness. Furthermore, the ESP32 uses its built-in WIFI capability to transmit data to the Thing Speak cloud platform, enabling remote monitoring, graphical visualization, and data storage. This system is designed to be cost-effective, portable, and user-friendly, providing an efficient solution for continuous health monitoring and early detection of abnormal breathing patterns. In recent years, the integration of Internet of Things (IoT) technology in healthcare has significantly improved patient monitoring systems by enabling real-time data acquisition and remote accessibility. IoT-based solutions allow continuous tracking of vital parameters without the need for hospital visits, thereby reducing healthcare costs and improving patient convenience. Especially in sleep-related disorders like sleep apnea, continuous monitoring plays a crucial role in early diagnosis and prevention of severe complications. The proposed system leverages embedded systems and wireless communication technologies to create an efficient and reliable health monitoring solution. By using ESP32, which combines processing capability with built-in WIFI, the system ensures seamless data transmission to cloud platforms. The use of Thing Speak enables visualization of health data in graphical formats, making it easier for users and healthcare professionals to analyze trends and detect abnormalities.

1.1 Overview

Sleep Apnea is a critical health condition that affects a large number of people worldwide, often remaining undiagnosed due to the lack of continuous monitoring facilities. It is characterized by repeated pauses in breathing during sleep, which can lead to reduced oxygen supply to vital organs. Early detection and continuous monitoring are essential to prevent severe health complications. However, existing monitoring systems are mostly hospital-based, expensive, and not suitable for long-term home use. In this context, the proposed system provides a smart and portable solution for monitoring sleep apnea using IoT. The system is built using an ESP32 microcontroller integrated with an IR sensor to detect physiological signals such as heart rate and oxygen saturation levels. The collected data is processed using a threshold-based algorithm and displayed locally on an LCD screen. Furthermore, the system utilizes the built-in WiFi capability of ESP32 to transmit the collected data to the Thing Speak cloud platform. This allows users and healthcare providers to monitor the patient's condition remotely through graphical visualization and data analysis. The system also supports data storage and download features, making it useful for long-term monitoring and medical evaluation. Overall, this project provides a simple, cost-effective, and efficient solution for real-time sleep apnea monitoring in home healthcare environments.

1.2 Problem Statement

Sleep Apnea is a serious and often undiagnosed sleep disorder that causes repeated interruptions in breathing during sleep. Despite its severe health implications, including cardiovascular diseases, hypertension, and fatigue, many individuals remain unaware of their condition due to the lack of accessible and continuous monitoring systems. Early detection is crucial, but existing diagnostic methods are not convenient for regular use. The current standard method for diagnosing sleep apnea is Polysomnography (PSG), which requires patients to stay overnight in specialized hospital laboratories. This process involves complex equipment, multiple sensors, and continuous medical supervision, making it expensive, time-consuming, and uncomfortable for patients. Additionally, such clinical setups do not reflect natural sleeping conditions, which may affect the accuracy of diagnosis. Therefore, there is a need for a cost-effective, portable, and easy-to-use system that can continuously monitor sleep patterns in a home environment. The system should be capable of detecting abnormal breathing conditions in real time and providing immediate feedback to the user.

This project addresses these challenges by proposing an IoT-based solution that enables continuous monitoring, remote data access, and early detection of sleep apnea, thereby improving healthcare accessibility and patient safety. In addition, the increasing demand for smart healthcare solutions highlights the importance of integrating modern technologies such as IoT into medical systems. A real-time monitoring system that can operate continuously without human intervention is essential for improving patient outcomes. By providing instant data visualization and remote access through cloud platforms, the proposed system ensures better tracking of health conditions and supports timely medical decisions, reducing the risks associated with delayed diagnosis.

1.3 Project Goal and Objectives

The primary goal of this project is to design and develop a Portable IoT-Based Sleep Apnea Monitoring System that enables continuous and real-time monitoring of physiological parameters in a home healthcare environment. The system aims to provide a cost-effective, non-invasive, and user-friendly solution for early detection of sleep apnea conditions, thereby reducing the dependency on complex and expensive hospital-based diagnostic methods such as Polysomnography (PSG). To achieve this goal, several specific objectives have been identified. First, to develop a reliable hardware system using an ESP32 microcontroller integrated with an IR sensor for capturing physiological signals such as heart rate and oxygen levels. Second, to implement a threshold-based algorithm using embedded C in the Arduino IDE for real-time data processing and detection of abnormal conditions. Third, to design a local display system using an LCD module for instant user feedback. Fourth, to establish wireless communication using the built-in WIFI module of ESP32 for transmitting data to the Thing Speak cloud platform. Fifth, to enable remote monitoring through graphical visualization and support data storage and download features for further analysis. Additionally, the project focuses on developing a reliable and scalable system that can operate continuously with minimal human intervention. The system is designed to ensure stable data acquisition, efficient processing, and seamless communication between hardware and cloud platforms. It also provides a foundation for future enhancements such as integration of additional sensors, mobile applications, and advanced analytics techniques, making it adaptable for broader healthcare monitoring applications. Overall, the project aims to provide an efficient, portable, and scalable healthcare monitoring solution that enhances early diagnosis, improves patient safety, and demonstrates the practical application of IoT technology in modern healthcare systems.

1.4 Project Scope

The scope of this project focuses on the design, development, and implementation of a Portable IoT-Based Sleep Apnea Monitoring System intended for home healthcare applications. The system aims to provide continuous and real-time monitoring of physiological parameters such as heart rate and oxygen levels using an IR sensor integrated with an ESP32 microcontroller. By leveraging embedded systems and wireless communication technologies, the project provides a non-invasive and user-friendly solution that can be easily deployed in a home environment without requiring specialized medical infrastructure.

The project includes the development of both hardware and software components to achieve reliable monitoring. On the hardware side, the system consists of an ESP32 microcontroller, IR sensor, LCD display, and power supply unit, all integrated to form a compact and portable device. On the software side, the system is implemented using embedded C programming in the Arduino IDE, where a threshold-based algorithm is used to process sensor data and detect abnormal conditions related to sleep apnea. The processed data is displayed locally on the LCD for real-time feedback and is also transmitted to the Thing Speak cloud platform using the ESP32's built-in WIFI module.

However, the scope of this project is limited to basic monitoring and detection using a threshold-based algorithm and does not include advanced machine learning techniques or clinical-grade diagnostic capabilities. The system is designed primarily as a preliminary screening tool rather than a complete replacement for traditional medical diagnostic systems such as Polysomnography (PSG). Additionally, the system currently focuses on a limited set of physiological parameters and does not incorporate multiple sensor fusion or advanced predictive analytics.

Despite these limitations, the project provides a strong foundation for future enhancements. The system can be extended by integrating additional sensors such as SpO_2 modules, implementing machine learning algorithms for improved accuracy, and developing mobile or web-based applications for enhanced user interaction. Overall, the scope of this project demonstrates the feasibility and effectiveness of using IoT technology for real-time health monitoring and highlights its potential to improve accessibility, affordability, and efficiency in modern healthcare systems.

CHAPTER 2

LITERATURE REVIEW

2.1 RESEARCH PAPER 1

Topic:

U. R. Acharya, S. V. Sree, G. Swapna, R. J. Martis, and J. S. Suri, "Automated diagnosis of sleep apnea using ECG signals: A review, "Knowledge-Based Systems, vol. 45, pp. 1–17, 2013.

Description:

This research paper presents a comprehensive review on the automated diagnosis of sleep apnea using ECG (Electrocardiogram) signals. The authors discussed how sleep apnea can be detected by analyzing physiological signals related to heart activity and respiratory variations. The paper mainly focuses on the use of signal processing techniques, feature extraction methods, and classification approaches for identifying sleep apnea conditions in an automated manner. The study highlights that sleep apnea is a serious sleep-related breathing disorder which can affect normal oxygen flow and cardiovascular functioning. Therefore, the paper emphasizes the importance of early and continuous monitoring using physiological parameters such as heart-related signals. This paper is useful for the proposed project because it provides a strong background on the importance of monitoring physiological signals for sleep apnea-related analysis.

2.2 RESEARCH PAPER 2

Topic:

T. Penzel, J. W. Kantelhardt, L. Grote, J. H. Peter, and A. Bunde, "Comparison of detrended fluctuation analysis and spectral analysis for heart rate variability in sleep Apnea," IEEE Transactions on Biomedical Engineering, vol. 50, no. 10, pp. 1143–1151, 2003.

Description:

This research paper focuses on the analysis of heart rate variability (HRV) in patients affected by sleep apnea. The authors compared two important analytical methods, namely Detrended Fluctuation Analysis (DFA) and Spectral Analysis, to study the changes in heart rate patterns during sleep-related breathing disturbances. The study explains that sleep apnea can influence the normal rhythm of the heart due to repeated interruptions in breathing during sleep. Because of this, the heart rate pattern may show noticeable fluctuations, which can be used as an indicator for identifying sleep apnea-related abnormalities.

In this paper, the authors investigated how different signal analysis techniques can be applied to evaluate heart rate variability and understand the physiological impact of sleep apnea. The paper demonstrates that heart-related signal variations contain meaningful information which can support monitoring and analysis of sleep disorders. The comparison made in this study is important because it shows that physiological parameters such as heart rate are valuable for sleep apnea observation. The paper also provides insight into how biomedical signal analysis can be used in healthcare monitoring systems.

2.3 RESEARCH PAPER 3

Topic:

S. Behar, J. Oster, Q. Li, and G. D. Clifford, "ECG signal quality during arrhythmia and its application to sleep apnea detection," IEEE Journal of Biomedical and Health Informatics, vol. 18, no. 3, pp. 876–885, 2014.

Description:

This research paper discusses the importance of ECG signal quality in biomedical monitoring and explains its application in the detection of sleep apnea. The authors focused on how the quality of ECG signals can affect the accuracy of physiological analysis, especially in conditions where heart rhythm variations and abnormalities are present. This research paper discusses the importance of ECG signal quality in biomedical monitoring and explains its application in the detection of sleep apnea. The authors focused on how the quality of ECG signals can affect the accuracy of physiological analysis, especially in conditions where heart rhythm variations and abnormalities are present. The authors examined methods for evaluating ECG signal quality and discussed how reliable physiological signal acquisition is necessary for accurate health monitoring. The paper also demonstrates that ECG-based analysis can be applied not only for cardiac monitoring but also for detecting abnormalities associated with sleep apnea.

CHAPTER 3

SYSTEM DESIGN & ANALYSIS

3.1 Analysis of Existing System

The diagnosis and monitoring of sleep apnea in existing healthcare systems are generally carried out using hospital-based clinical equipment and specialized sleep study methods such as Polysomnography (PSG). These traditional systems are considered accurate for medical diagnosis, but they also come with several practical limitations when it comes to continuous patient monitoring and home healthcare support. Sleep apnea is a condition that often requires long-term observation, especially during sleep, and this makes the use of only hospital-based systems inconvenient for many patients. In the conventional monitoring method, the patient is usually required to stay overnight in a sleep laboratory or hospital under the supervision of medical staff. During this process, multiple sensors and wires are attached to the patient's body in order to monitor breathing patterns, oxygen levels, heart rate, and other physiological signals. Although this setup provides detailed information, it can cause discomfort to the patient and may disturb natural sleep. As a result, the collected data may not always reflect the patient's normal sleeping condition at home.

Another major limitation of the existing system is its high cost. Sleep studies conducted in hospitals require expensive diagnostic equipment, technical support, and specialist supervision. This makes it difficult for many individuals, especially people living in rural or low-resource areas, to access proper sleep monitoring facilities. In addition, patients with mild or early symptoms may ignore testing because of the inconvenience and cost involved. Existing systems also lack real-time remote accessibility for home users. In many cases, the patient's data is not continuously available for personal tracking or easy monitoring outside the hospital environment. This creates a gap between clinical diagnosis and daily health observation. Since sleep apnea may be associated with variations in heart rate and blood oxygen saturation (SpO_2), there is a strong need for a system that can monitor these parameters continuously in a simpler and more affordable way. From this analysis, it is clear that the existing systems are effective in clinical environments but are not ideal for regular, user-friendly, home-based monitoring.

These limitations motivated the development of the proposed IoT-Based Sleep Apnea System.

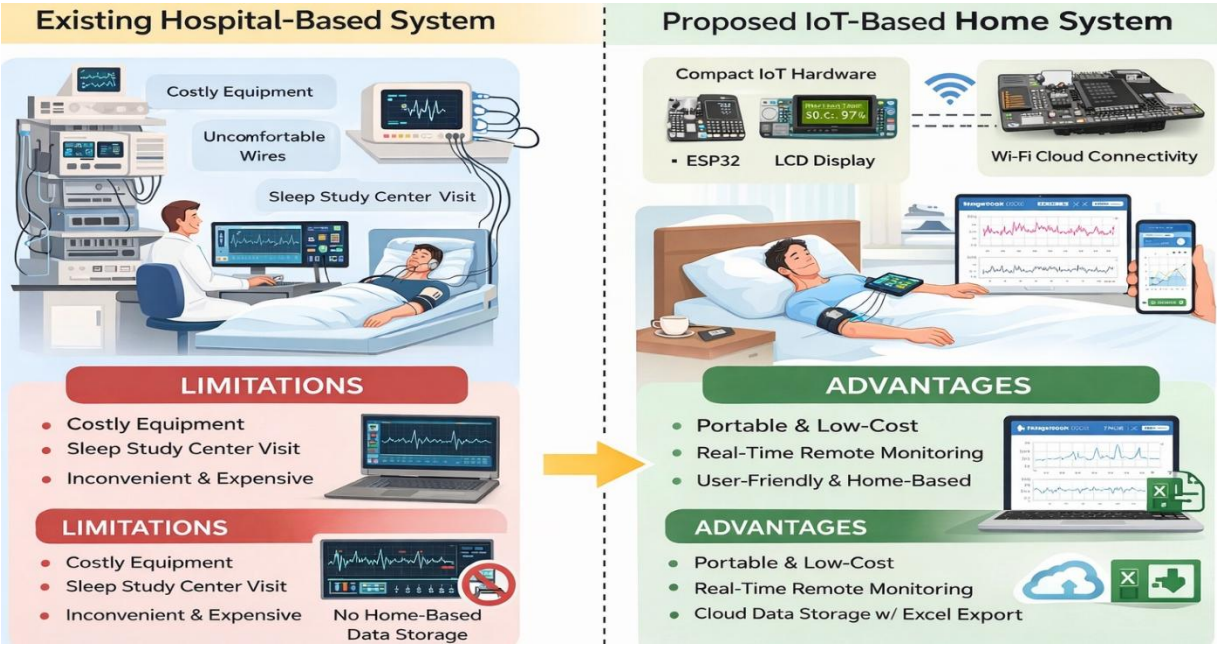

Figure 3.1 Existing System and Proposed System Comparison

The above figure 3.1 shows the difference between the traditional hospital-based sleep monitoring system and the proposed IoT-based home healthcare monitoring system. It highlights the limitations of the existing system and the advantages of the proposed portable monitoring approach.

3.2 Proposed System Architecture

The proposed system architecture of the Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare is designed to provide a simple, low-cost, and effective solution for monitoring important physiological parameters related to sleep apnea. The system is mainly developed to observe the user's heart rate and oxygen saturation (SpO_2) in real time and make the readings available both locally and remotely through cloud technology. The architecture of the system consists of several interconnected components, namely the IR Sensor, ESP32 microcontroller, LCD display, Power Supply Unit, WiFi communication, Thing Speak cloud platform, and the user monitoring interface. Each component plays an important role in the data collection, processing, display, and transmission stages of the system. The IR Sensor acts as the primary sensing component in the system. It is responsible for detecting physiological signals from the user's body, which are then used to estimate parameters such as heart rate and oxygen-related measurements. These sensed values are sent to the ESP32 microcontroller, which acts as the main processing unit of the system.

The ESP32 receives the input data from the sensor, processes it, and prepares it for display and transmission. Since the ESP32 has built-in WiFi capability, it can also send the processed data directly to the cloud platform without requiring an additional external communication module. This makes the system compact and suitable for portable healthcare applications. The processed values are displayed on the LCD display, allowing the user to view real-time readings such as Heart Rate and SpO_2 instantly. This local display is useful for immediate observation without needing internet access every time. For remote monitoring, the ESP32 transmits the collected data to the Thing Speak cloud platform using WiFi connectivity. Once uploaded, the data is stored in the cloud and represented in the form of graphs, such as Heart Rate Graph and SpO_2 Graph. This enables the user to analyze health trends over time.

Another useful feature of the system is the ability to download the stored data in Excel format which can be used for record keeping, analysis, or future healthcare reference. This makes the system more practical and beneficial for both home users an Overall, Fig 3.2 the proposed architecture is designed to create a complete health monitoring loop that starts from data sensing and ends with cloud visualization and user access. It provides a user-friendly, portable, and IoT-enabled solution for home healthcare monitoring's caregivers.

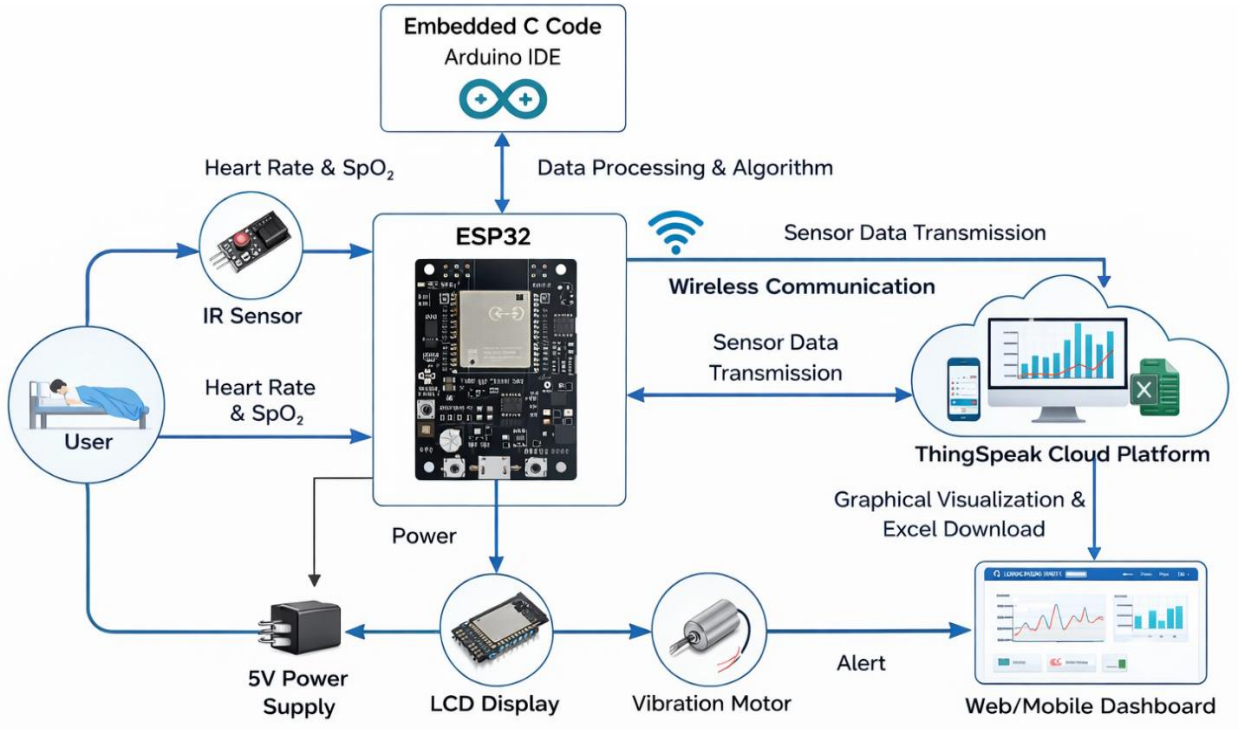

Fig 3.2 Overall Architecture Diagram of the Proposed System

3.3 System Requirements

The successful implementation of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare requires both hardware components and software tools. These system requirements are essential for collecting physiological data, processing the readings, displaying the output, and transmitting the monitored values to the cloud platform. The system is designed to be simple, compact, and cost-effective. Therefore, the selected components are chosen based on ease of use, availability, compatibility, and suitability for real-time health monitoring applications. The hardware requirements include sensing, processing, display, power, and communication elements. These physical components are directly involved in building the monitoring device. The main hardware components used in the system are IR Sensor, ESP32 microcontroller, LCD display, and Power Supply Unit. These components work together to detect physiological signals, process the values, and display the output. The software requirements include the programming and communication tools needed to operate the system. The software side of the project is responsible for coding, uploading, monitoring, and cloud data visualization. The major software tools used in this project are Arduino IDE, WiFi communication setup, and ThingSpeak cloud platform. These tools help in developing the embedded program, sending sensor data to the internet, and displaying the monitored values in graphical format. The proper combination of hardware and software requirements ensures that the system works efficiently in a home healthcare environment. These requirements also make the project practical for educational, healthcare, and IoT-based monitoring purposes.

3.4 Hardware Components

The hardware components used in the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare play a major role in collecting, processing, displaying, and transmitting the physiological data of the user. The system is designed using compact, affordable, and easily available electronic components so that it can be effectively implemented for home healthcare monitoring. Each hardware component in the system has a specific function. The IR sensor is used for sensing physiological signals, the ESP32 microcontroller acts as the processing and communication unit, the LCD display is used for displaying real-time values, and the Power Supply Unit provides the necessary electrical power for system operation. Together, these hardware components form the core physical structure of the proposed monitoring system. The following subsections describe each hardware component used in the project in detail.

3.4.1 IR Sensor

The IR (Infrared) Sensor is one of the most important sensing components used in the proposed sleep apnea monitoring system. It is mainly used to detect physiological signals from the human body, especially for monitoring parameters such as heart rate and oxygen-related changes. In this project, the IR sensor acts as the primary input device that collects the user's body-related signal for further processing.

The working principle of the IR sensor is based on the reflection or absorption of infrared light. When the sensor is placed on or near the human body, it emits infrared light and receives the reflected signal from the tissue or blood flow.

Variations in blood circulation and pulse movement cause small changes in the reflected signal, which can then be analyzed to estimate heart rate and related physiological conditions in the context of sleep apnea monitoring, continuous observation of heart rate and oxygen-related physiological variations is important because abnormal breathing patterns during sleep may influence these health parameters. Therefore, the IR sensor is used to continuously collect the required body signal and send it to the ESP32 microcontroller for further processing. The sensor is selected in this project because it is compact, low-cost, easy to interface, and suitable for real-time monitoring applications. It is also useful for portable health monitoring systems due to its simple design and efficient sensing capability. Thus, the IR sensor serves as the first stage of the proposed monitoring system by capturing the user's physiological signal and initiating the data acquisition process.

Fig 3.3 IR Sensor

The above figure 3.3 shows the IR sensor used in the proposed monitoring system. It is responsible for sensing physiological signals from the user's body.

3.4.2 ESP32 Microcontroller

The ESP32 microcontroller is the central processing unit of the proposed IoT-Based Sleep Apnea Monitoring System. It acts as the brain of the entire system by receiving sensor data, processing the values, controlling the display unit, and transmitting the monitored data to the cloud platform through WiFi communication. The ESP32 is chosen for this project because it is a powerful and compact microcontroller with built-in WiFi and Bluetooth capability, making it highly suitable for IoT-based healthcare applications. Unlike traditional microcontrollers that require external communication modules, the ESP32 can directly connect to wireless networks, which simplifies the circuit design and reduces overall system complexity. In this project, the ESP32 receives the physiological signal data from the IR sensor and processes it to derive useful health parameters such as heart rate and SpO_2. After processing, the microcontroller sends the real-time values to the LCD display so that the user can instantly observe the readings. In addition to local display functionality, the ESP32 also transmits the monitored data to the Thing Speak cloud platform using its inbuilt WiFi feature. This enables real-time remote monitoring, cloud storage, graphical visualization, and data analysis. Thus, the ESP32 plays a dual role in the system by acting as both a processing unit and a communication unit Another advantage of using ESP32 is its low power consumption, fast processing speed, and compatibility with Arduino IDE, which makes programming and implementation easier. Due to these features, the ESP32 is widely used in smart healthcare, wearable devices, and real-time monitoring systems. Hence, the ESP32 microcontroller forms the core control unit of the proposed system and ensures efficient integration between sensing, processing, display, and cloud communication.

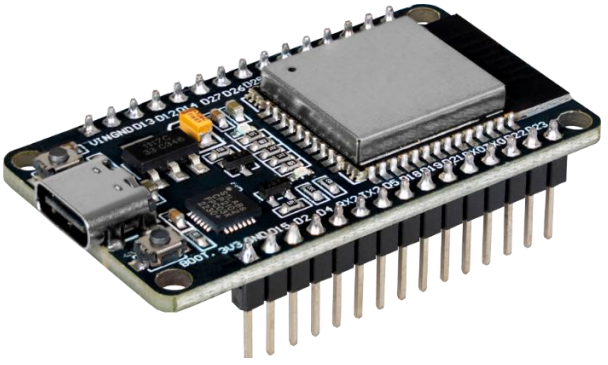

Fig 3.4 ESP32 Microcontroller

The above figure 3.4 shows the ESP32 microcontroller used in the proposed system. It acts as the main control and processing unit by receiving sensor data, processing the readings, displaying the values, and transmitting the data to the cloud platform.

3.4.3 LCD Display

The LCD (Liquid Crystal Display) is used in the proposed system to provide a real-time visual output of the monitored physiological parameters. It acts as the local display unit that allows the user to directly observe important health values such as heart rate and SpO_2 without requiring a mobile device or internet connection.

In this project, the LCD display is connected to the ESP32 microcontroller, which continuously updates the screen with the processed sensor readings. This makes it possible for the user to instantly view the health data while the system is operating. The LCD display improves the usability of the system by providing immediate feedback in a simple and readable format.

The use of an LCD display is important in healthcare monitoring systems because it allows quick verification of whether the system is functioning correctly and whether the physiological values are within a normal range. It is especially useful in-home healthcare applications, where users may prefer a direct display instead of depending only on cloud dashboards. The LCD used in this project is selected because it is lightweight, easy to interface, low power consuming, and suitable for embedded systems. It also enhances the portability of the system and makes the monitoring device more user-friendly. It is especially useful in-home healthcare monitoring because it gives immediate visual feedback. The LCD display can also be used to show system status messages such as WiFi connection status or data transmission confirmation. Furthermore, the use of an LCD display improves the overall usability and accessibility of the system. It reduces dependency on external devices like mobile phones or computers for basic monitoring. Hence, it plays a crucial role in making the system more user-friendly, reliable, and efficient.

Fig 3.5 LCD Display

The above figure 3.5 shows the LCD display used in the proposed monitoring system. It displays the real-time values of heart rate and SpO_2, allowing the user to directly observe the monitored health parameters.

3.4.4 Power Supply Unit:

The Power Supply Unit is an essential hardware component in the proposed system, as it provides the required electrical power for the proper functioning of all electronic modules. In the Portable IoT-Based Sleep Apnea Monitoring System, the power supply ensures that the ESP32, IR sensor, LCD display, and other connected components operate continuously and reliably. A stable and regulated power source is important in healthcare monitoring systems because fluctuations in power may affect sensor readings, data processing, or communication performance. Therefore, an appropriate power supply is used in this project to maintain system stability and uninterrupted monitoring.

In this system, the power supply may be provided using a 5V adapter, USB power source, or a portable battery source, depending on the implementation setup. Since the project is intended for portable home healthcare use, the power arrangement is designed to be simple and convenient for regular operation. The power supply unit also supports the portability of the system by allowing the device to be operated in different home environments. It helps in maintaining reliable performance during real-time monitoring and data transmission activities. Overall, the WIFI module enables seamless communication between the hardware system and the cloud platform, making real-time monitoring, data storage, and remote access possible in the proposed system. Additionally, the power supply unit supports long-term monitoring by providing consistent performance. It can be designed using batteries or regulated power sources depending on the application requirements. Proper power management also improves the efficiency and lifespan of the system the power supply unit plays a vital role in ensuring the stability, reliability, and continuous operation of the sleep Apnea monitoring system.

Fig 3.6 Power Supply Unit

The above figure 3.6 shows the power supply unit used in the proposed system. It provides the necessary electrical power to the ESP32, sensor, and display components for continuous and stable operation.

3.4.5 Vibration Motor:

The vibration motor is used in the proposed system as an alert mechanism to notify the user during abnormal conditions. It is a small DC motor that produces vibration when electrical power is supplied. This component plays an important role in ensuring user safety by providing immediate physical alerts. In this project, the vibration motor is activated when abnormal physiological conditions such as irregular heart rate or possible sleep Apnea events are detected. The ESP32 microcontroller controls the operation of the motor based on the processed sensor data. When the detected values exceed predefined threshold limits, the motor is triggered to alert the user. The vibration motor is typically interfaced with the ESP32 using a transistor-based switching circuit to handle the required current. It is compact, low-cost, and consumes less power, making it suitable for portable healthcare devices. Overall, fig 3.7 the vibration motor enhances the system by providing real-time alerts, ensuring that the user is immediately informed about critical health conditions, even without looking at the display.

Fig 3.7 Vibration Motor

3.5 Software &Communication:

The software and communication components used in the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare are responsible for programming, controlling, data transmission, cloud storage, and remote monitoring of physiological parameters. While the hardware components perform sensing and display functions, the software and communication tools ensure that the collected data is processed, uploaded, and visualized efficiently. In this project, the major software and communication components include Arduino IDE, WiFi communication, and the Thing Speak cloud platform.

These components play an important role in the successful implementation of the system by enabling real-time data acquisition, wireless communication, cloud-based storage, and graphical representation of monitored values. The following subsections explain the software and communication tools used in this project in detail.

3.5.1 Arduino IDE:

The Arduino IDE (Integrated Development Environment) is the primary software tool used for writing, editing, compiling, and uploading the program code to the ESP32 microcontroller in the proposed system. It provides a simple and user-friendly platform for embedded system programming, making it highly suitable for IoT and healthcare-based applications. In this project, the Arduino IDE is used to develop the Embedded C code required for reading the sensor values, processing physiological data, displaying the output on the LCD, and sending the monitored values to the cloud platform. It acts as the software environment in which the entire operational logic of the system is implemented. One of the major advantages of the Arduino IDE is its simplicity and compatibility with a wide range of microcontrollers, including the ESP32. It supports the use of multiple libraries for sensor interfacing, LCD communication, WiFi connectivity, and cloud integration. This reduces programming complexity and helps in faster development of the monitoring system. The Arduino IDE also allows easy debugging and testing of the code, which is important during hardware implementation and system integration. Through this software, the user can verify whether the sensor readings are correctly obtained, whether the LCD is displaying the right values, and whether the ESP32 is successfully transmitting data to the cloud.

Thus, the Arduino IDE acts as the main software development tool for the proposed system and provides the programming support required for the proper operation of the device

Fig 3.8 Arduino IDE

The above figure 3.8 shows the Arduino IDE software used for developing and uploading the embedded program to the ESP32 microcontroller. It is used for coding, debugging, and controlling the operation of the proposed monitoring system.

3.5.2 WiFi Communication:

WiFi communication is an essential part of the proposed IoT-Based Sleep Apnea Monitoring System, as it enables wireless transmission of physiological data from the ESP32 microcontroller to the Thing Speak cloud platform. This communication feature allows the system to perform real-time remote monitoring and cloud-based data storage. In this project, the ESP32 uses its built-in WiFi capability to connect to an available wireless network. Once connected, the microcontroller transmits the monitored data such as heart rate and SpO_2 values to the cloud platform at regular intervals. This wireless communication eliminates the need for physical wired connections between the monitoring device and the user interface.

The use of WiFi communication improves the practicality and portability of the system. It allows the user to monitor health-related data from a mobile phone, computer, or cloud dashboard without being physically close to the device. This is especially useful in-home healthcare applications, where convenience and remote accessibility are important. Another major benefit of WiFi communication is that it supports continuous data updates, which helps in real-time observation of the user's physiological condition. It also makes the system more scalable, as future improvements such as mobile notifications, alert systems, and doctor dashboards can be easily integrated.

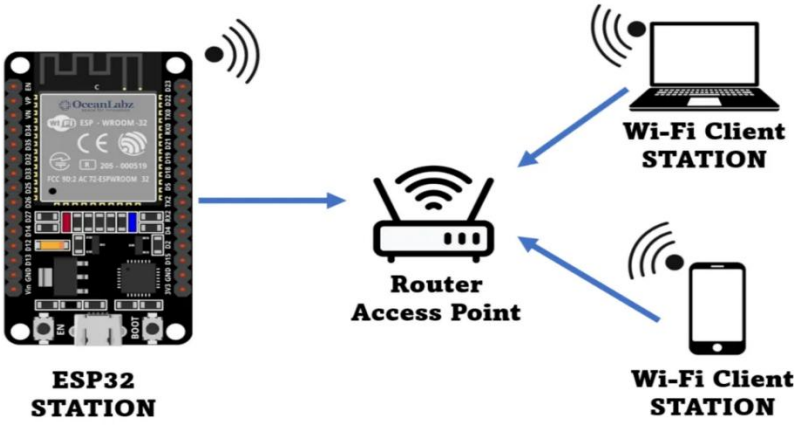

Fig 3.9 WiFi Communication

The above figure 3.9 represents the WiFi communication process used in the proposed system. It enables the ESP32 microcontroller to wirelessly transmit the monitored physiological data to the cloud platform for remote access and analysis.

3.5.3 Thing Speak Cloud Platform

The Thing Speak Cloud Platform is used in the proposed system for cloud-based data storage, graphical visualization, and remote monitoring of the physiological parameters collected from the user. It acts as the online platform where the monitored values such as heart rate and SpO_2 are uploaded, stored, and displayed in an organized manner, the ESP32 microcontroller sends the processed sensor readings to the Thing Speak cloud platform through WiFi communication. Once the data reaches the cloud, it is automatically recorded and presented in the form of graphs and field-based data entries. This helps in observing the variation of health parameters over time and makes the monitoring process more meaningful and user-friendly. Thing Speak is selected for this project because it is easy to configure, compatible with IoT devices, and supports real-time data updates. It also provides features such as data logging, graph plotting, and Excel-compatible data export, which are useful for report preparation, long-term health observation, and result analysis. The platform can be accessed through a web browser or compatible device, which enables the user to view the monitored data remotely. This makes the system highly suitable for home healthcare monitoring, where the user may need to check health trends without device.

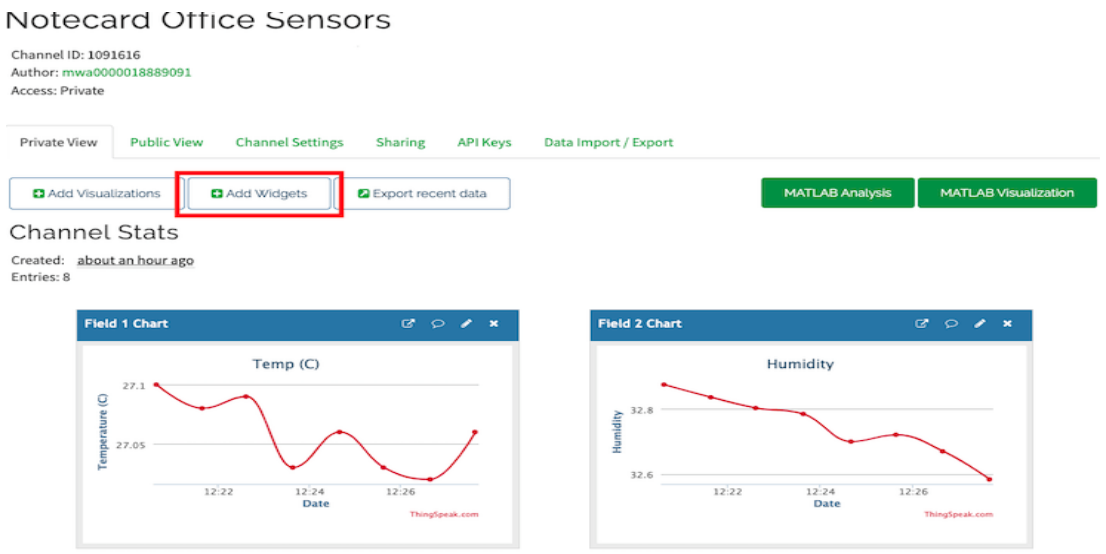

Fig 3.10 Thing Speak Cloud Platform

The above figure 3.10 shows the Thing Speak cloud platform used in the proposed system. It is used for storing, visualizing, and monitoring the physiological data transmitted by the ESP32, enabling real-time graphical analysis and remote health observation.

3.6 Data Flow Diagram and Block Diagram

The Data Flow Diagram (DFD) and Block Diagram are important design representations used in the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare. These diagrams help in understanding how the system works internally by showing the movement of data and the relationship between different hardware and software components. The Data Flow Diagram is used to represent the flow of physiological data from the user through the sensing, processing, transmission, and output stages. It clearly illustrates how the input signal is collected, processed, displayed, and stored in the cloud platform for remote monitoring. The Block Diagram, on the other hand, provides a simplified structural view of the system by representing the major components as interconnected functional blocks. It helps in understanding the overall system architecture in a technical and organized manner. Both diagrams are essential in system design because they provide a clear visual representation of how the proposed monitoring system operates. They also help in explaining the project effectively during report evaluation and viva presentation.

3.6.1 Data Flow Diagram

The Data Flow Diagram (DFD) of the proposed system represents the movement of physiological data from the user to the monitoring outputs. It describes how the input signal is collected, processed, transmitted, and finally presented to the user in both local and cloud-based forms. In the proposed system, the user acts as the source of physiological input, particularly heart rate and SpO_2-related body signals. These signals are first sensed by the IR sensor, which captures the necessary data from the user's body and sends it to the ESP32 microcontroller. The ESP32 acts as the central processing unit in the data flow process.

It receives the sensor input, processes the physiological values, and generates meaningful output such as heart rate and SpO_2 readings. These processed values are then sent to two major output paths. The first output path is the LCD display, where the real-time readings are shown directly to the user. This provides immediate access to the monitored values without requiring internet connectivity.

The second output path is the WiFi communication channel, through which the ESP32 transmits the processed data to the Thing Speak cloud platform. Once uploaded to the cloud, the data is stored and visualized in the form of graphs and downloadable records. This enables remote monitoring through mobile or web interfaces.

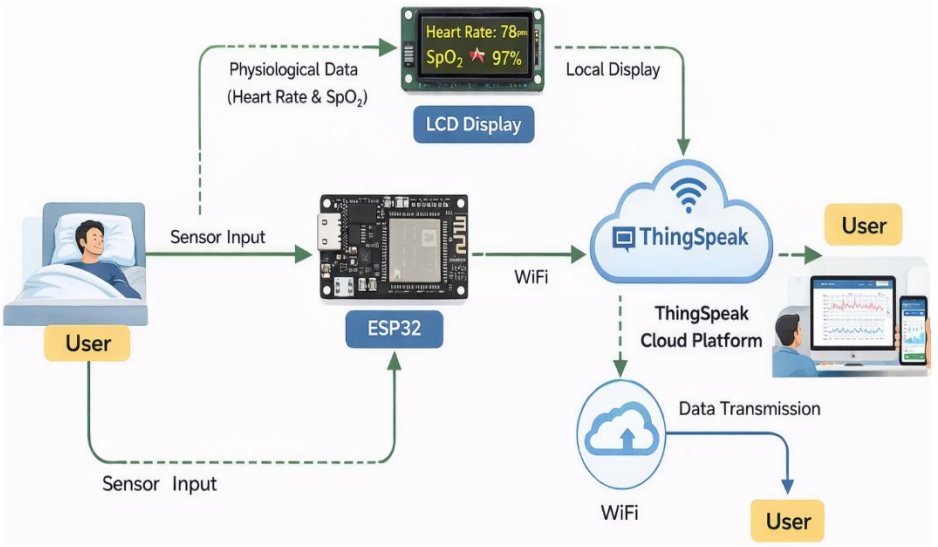

Figure 3.11 Data Flow Diagram

The above figure 3.11 represents the data flow diagram of the proposed monitoring system. It illustrates how physiological data is collected from the user, processed by the ESP32, displayed locally on the LCD, and transmitted to the cloud platform for remote monitoring and storage.

3.6.2 Block Diagram

The Block Diagram of the proposed Portable IoT-Based Sleep Apnea Monitoring System provides a simplified and structured representation of the major functional units involved in the system. It helps in understanding the technical arrangement and interaction between the input, processing, communication, display, and cloud modules.

In the IR sensor is represented as the input block, which is responsible for sensing the physiological parameters from the user. The sensed data is then transferred to the ESP32 microcontroller, which acts as the central control and processing block of the system. The ESP32 performs signal processing and sends the output to the LCD display, which serves as the local output unit. At the same time, the ESP32 also uses WiFi communication to transmit the monitored data to the Thing Speak cloud platform.

The Thing Speak cloud platform functions as the cloud storage and data visualization block, where the uploaded values are represented as heart rate graphs, SpO_2 graphs, and downloadable records such as Excel files. The user can then access the output through a mobile or web dashboard, which acts as the final monitoring interface. Fig 3.12

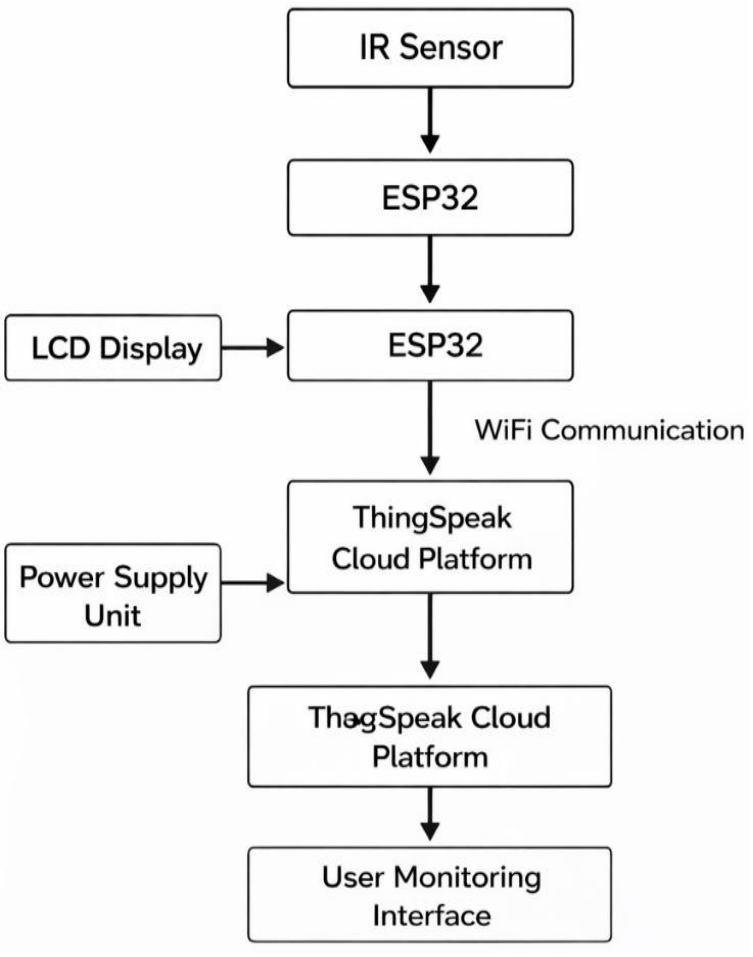

Fig 3.12 Block Diagram

3.7 Working principle

The working principle of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare is based on the continuous sensing, processing, display, transmission, and monitoring of physiological parameters related to the user's sleep condition. The system is mainly designed to monitor heart rate and SpO_2 (oxygen saturation) values in real time and make them available both locally and remotely for observation. The working process begins when the IR sensor is used to sense physiological signals from the user's body.

The sensor detects variations related to pulse activity and oxygen-related physiological changes. These sensed values act as the raw input data for the monitoring system. The collected sensor data is then sent to the ESP32 microcontroller, which acts as the main control and processing unit of the system.

The ESP32 receives the input signal from the sensor and processes it using the programmed embedded logic. Based on the incoming signal, the microcontroller calculates and prepares the health-related output values such as heart rate and SpO_2. After processing the data, the ESP32 sends the calculated values to the LCD display, where the user can directly observe the real-time readings. This local display helps the user to instantly check the monitored physiological parameters without requiring any additional external device. At the same time, the ESP32 uses its built-in WiFi communication feature to establish a wireless connection with the internet.

Through this communication channel, the monitored values are transmitted to the Thing Speak cloud platform. The cloud platform stores the incoming data and displays it in the form of graphs and recorded data fields. Using Thing Speak, the user can observe the variation of heart rate and SpO_2 values over time. The platform also allows the monitored data to be downloaded in Excel format, which can be useful for further analysis, record maintenance, and healthcare reference The final output of the system can be accessed through a mobile phone, computer, or other compatible device connected to the internet. This enables the user or caregiver to monitor the physiological parameters remotely in a convenient and user-friendly manner.

Overall, the working principle of the proposed system follows a simple but effective sequence: sensing → processing → display → wireless transmission → cloud monitoring → user

CHAPTER 4
METHODOLOGY

This chapter details the physical construction and the sequential software pipeline that transforms raw sensor data into actionable apnea predictions, thereby realizing the Sleep Apnea Monitoring System (SAMS).

4.1 Hardware and Software Implementation

The implementation of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare involves the integration of both hardware components and software tools to achieve real-time monitoring of physiological parameters. This phase focuses on the practical development of the system, including the setup of hardware modules, programming of the microcontroller, and configuration of communication with the cloud platform.

The hardware implementation includes the connection of the IR sensor, ESP32 microcontroller, LCD display, and power supply unit. Each component is carefully interfaced to ensure accurate data acquisition and stable system operation. The IR sensor is connected to the input pins of the ESP32 to capture physiological signals. The ESP32 processes the incoming data and sends the output to the LCD display, which shows real-time readings.

On the software side, the system is programmed using the Arduino IDE, where the embedded code is written to control sensor reading, data processing, display output, and cloud communication. The ESP32 is programmed to connect to a WiFi network and transmit the monitored values to the Thing Speak cloud platform. The implementation process also includes testing each module individually and then integrating all components to form a complete system. Proper debugging and verification are carried out to ensure that the system works efficiently and provides accurate results.

The implementation process also includes testing each module individually and then integrating all components to form a complete system. Proper debugging and verification are carried out to ensure that the system works efficiently and provides accurate results. Thus, the hardware and software implementation together form the foundation of the proposed monitoring system, enabling real-time data acquisition, processing, and remote monitoring.

4.1.1 Hardware Setup

The hardware setup of the proposed system involves assembling and connecting all the required electronic components in a proper and organized manner. The main components used in the hardware setup include the IR sensor, ESP32 microcontroller, LCD display, and power supply unit.

The IR sensor is connected to the ESP32 to capture physiological signals from the user. The ESP32 acts as the central controller and processes the incoming data. The LCD display is interfaced with the ESP32 to show real-time values such as heart rate and SpO_2. The power supply unit provides the required electrical energy for the proper functioning of all components.

All connections are made carefully using appropriate wiring to ensure stable operation. The hardware setup is designed to be compact and portable, making it suitable for home healthcare applications.

The proper arrangement of components is important to avoid signal interference, power issues, or data transmission errors. Once the hardware setup is completed, the system is tested to ensure that all components are functioning correctly

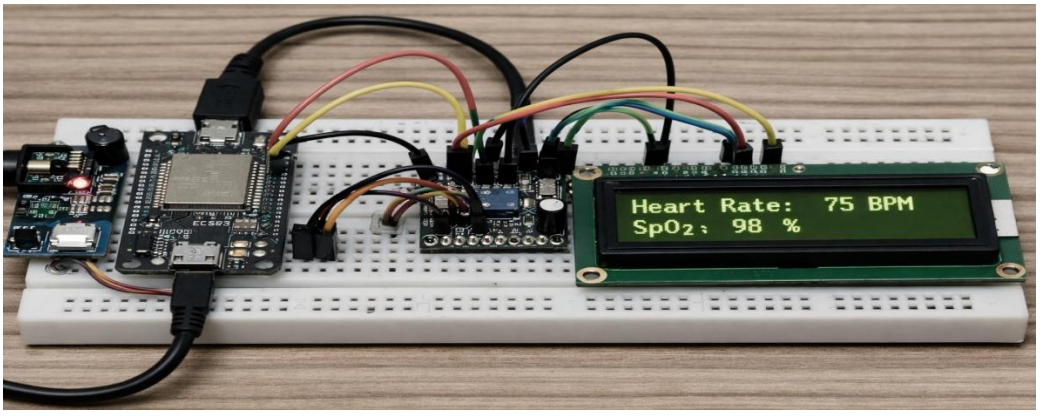

Fig 4.1 Hardware Setup

The above figure 4.1 shows the hardware setup of the proposed monitoring system. It includes the IR sensor, ESP32 microcontroller, LCD display, and power supply unit connected together to perform real-time physiological monitoring

4.1.2 Software Setup

The software setup of the proposed system involves programming the ESP32 microcontroller using the Arduino IDE and configuring the communication with the Thing Speak cloud platform. The software plays a crucial role in controlling the system operations and ensuring accurate data processing and transmission. The Arduino IDE is used to write the embedded C program that handles sensor data reading, data processing, LCD display output, and WiFi communication. Necessary libraries are included in the program to support sensor interfacing, display control, and internet connectivity. The ESP32 is configured with WiFi credentials to establish a connection with the internet. Once connected, the system sends the monitored data to the Thing Speak cloud platform using API keys and channel configuration. The data is then stored and visualized in graphical form. The software setup also includes testing and debugging the code to ensure that the system operates without errors. Proper validation is performed to check whether the sensor readings are accurate and whether the data is successfully uploaded to the cloud. Thus, the software setup enables the system to perform real-time monitoring, data transmission, and cloud-based visualization effectively.

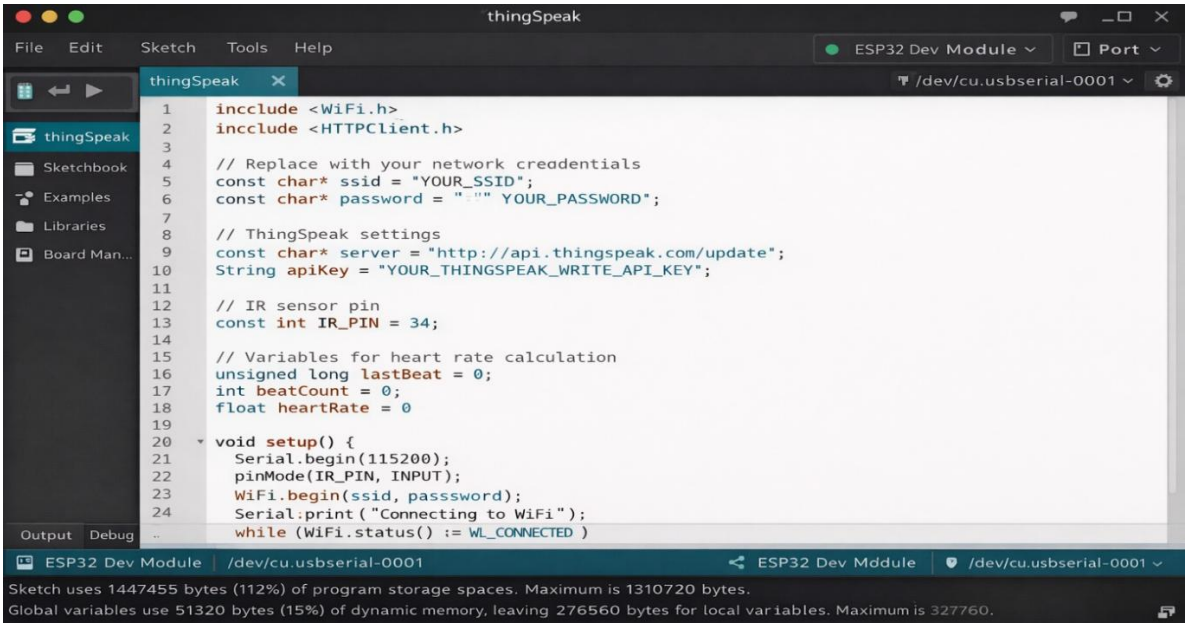

Fig 4.2 Software Setup

The above figure 4.2 shows the software setup of the proposed system, including the Arduino IDE programming environment and cloud configuration used for data transmission and monitoring.

4.2 Data Transmission

The data transmission protocol in the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare defines in this system, data transmission is achieved using WiFi communication and the HTTP protocol to send real-time sensor readings to the Thing Speak cloud platform. The ESP32 uses its built-in WiFi capability to connect to a wireless network using predefined credentials such as SSID and password. Once the ESP32 is connected to the WiFi network, it establishes communication with the Thing Speak server using the HTTP protocol. The HTTP protocol is a standard communication method used for transferring data over the internet. In this project, the ESP32 sends the data to the Thing Speak server by making an HTTP GET request. The data is transmitted using a unique API key, which is generated when a channel is created in Thing Speak. This API key ensures secure communication between the ESP32 device and the cloud platform. The transmitted data is included in the URL as parameters, such as field values representing heart rate and SpO_2. For example, the ESP32 constructs a URL in the following format:

http://api.thingspeak.com/update?api_key=YOUR_API_KEY&field1=value

When this request is sent, the Thing Speak server receives the data and updates the corresponding channel fields. The uploaded data is then stored in the cloud and can be visualized in the form of graphs or downloaded for further analysis. The data transmission process is performed at regular intervals to ensure continuous monitoring. Proper error handling is also included to check whether the data is successfully sent or if any communication failure occurs.

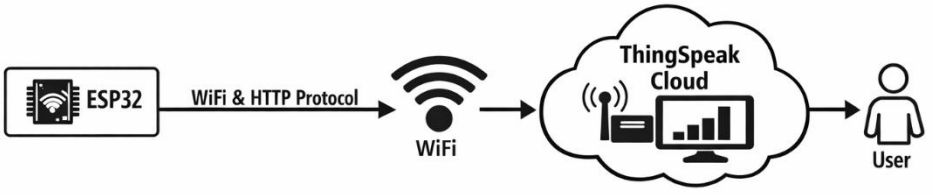

Fig 4.3 Data Transmission Process

The above figure 4.3 illustrates the data transmission process of the proposed system. It shows how the ESP32 microcontroller sends physiological data through WiFi using the HTTP protocol to the Thing Speak cloud platform for storage and visualization.

4.3 Software Implementation and Pipeline

The software implementation and pipeline of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare describe the sequence of operations performed by the system from sensor data acquisition to cloud-based output visualization. This section explains how the embedded program controls the entire monitoring process in a step-by-step manner. The software implementation is developed using Embedded C programming in the Arduino IDE, where the logic for sensor reading, data processing, output display, and cloud communication is written and uploaded to the ESP32 microcontroller. The software acts as the control mechanism that coordinates all functional stages of the proposed system. The working pipeline begins with the IR sensor, which continuously senses physiological signals from the user. These analog sensor values are read by the ESP32 through its input pins. Once the values are received, the software processes the incoming signal to identify pulse-related changes and estimate useful health parameters such as heart rate and SpO_2.

After processing the signal, the calculated values are sent to the LCD display, where the user can observe the readings in real time. At the same time, the software checks the WiFi connection status and prepares the data for cloud transmission. The next stage of the pipeline involves sending the processed data to the Thing Speak cloud platform using WiFi communication and HTTP requests. The uploaded values are stored in the cloud and represented graphically for further monitoring and analysis. This software pipeline ensures that the system performs continuous and organized monitoring of physiological parameters without manual intervention. It also improves the reliability of the proposed system by automating the process of sensing, calculation, display, and data transmission.

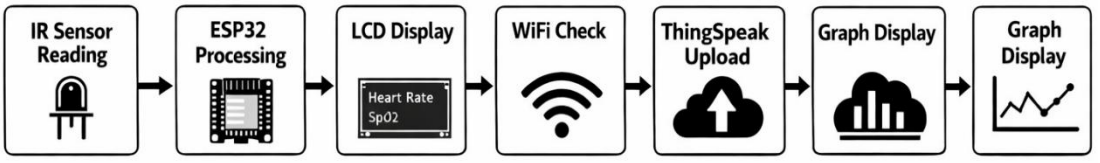

Fig 4.4 Software Implementation Pipeline

The above figure 4.4 illustrates the software implementation pipeline of the proposed monitoring system. It shows the step-by-step process from sensor data acquisition to ESP32 processing, LCD display, WiFi communication, and cloud-based data visualization.

4.3.1 Sensor Data Reading

The first stage in the software pipeline is sensor data reading, where the IR sensor continuously captures physiological signals from the user's body. These signals are converted into electrical values, which are then read by the ESP32 microcontroller through its analog input pin.

The embedded program is designed to repeatedly monitor the sensor values at regular intervals. The analog readings obtained from the sensor form the raw data used for further analysis and calculation. This step is essential because the accuracy of the final monitored values depends on the quality and consistency of the incoming sensor data. The software ensures that the sensor values are collected continuously and efficiently without missing important signal variations. This makes the system suitable for real-time physiological monitoring applications.

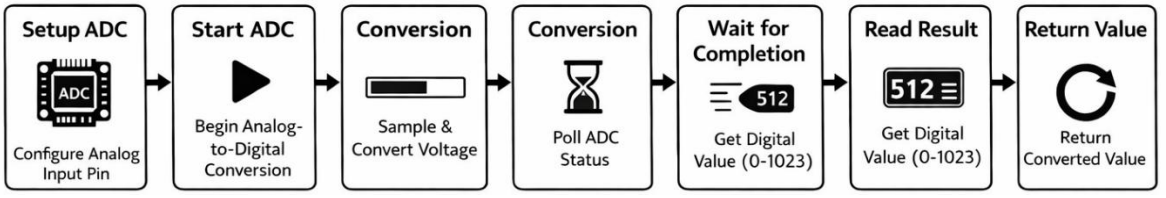

Fig 4.5 Sensor Data Reading Process

The above figure 4.5 represents the sensor data reading process in the proposed system. It shows how the IR sensor captures physiological signals and sends the raw analog values to the ESP32 microcontroller for further processing.

4.3.2 ESP32 Data Processing

Once the sensor data is collected, the next stage is data processing within the ESP32 microcontroller. In this stage, the raw sensor values are analyzed using the embedded logic written in the Arduino IDE. The software identifies pulse-related signal variations and uses them to estimate health-related parameters such as heart rate and SpO_2.

The ESP32 performs this task continuously in real time, making it possible for the system to update the monitored values regularly. The processing stage also includes simple threshold checking and timing calculations, which help in identifying valid pulse events and avoiding incorrect readings. This stage is highly important because it converts raw sensor data into meaningful physiological information that can be displayed and transmitted. Therefore, the ESP32 data processing unit acts as the intelligence center of the proposed system.

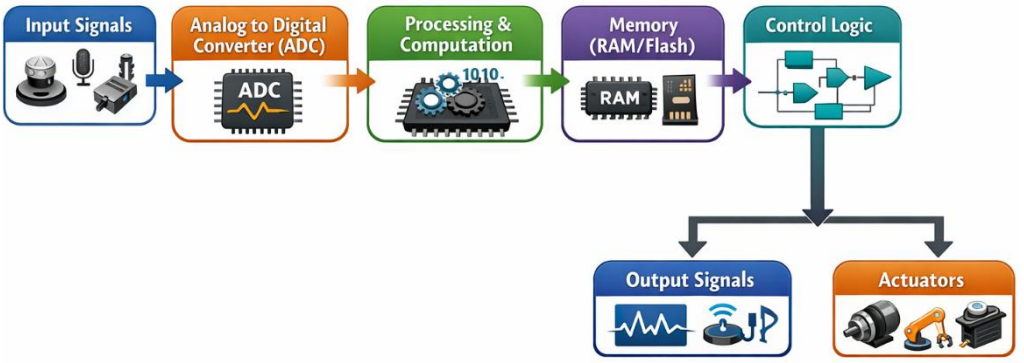

Fig 4.6 ESP32 Data Processing

The above figure 4.6 illustrates the data processing stage of the proposed system. It shows how the ESP32 microcontroller analyzes the raw sensor values and converts them into useful physiological parameters such as heart rate and SpO

4.3.3 LCD Output Display

After processing the physiological values, the system sends the results to the LCD display for local visualization. This stage allows the user to directly observe the real-time monitored values without depending on external devices or internet connectivity. The software is programmed to continuously update the LCD screen with the latest heart rate and SpO_2 readings. This improves the usability of the proposed system by providing immediate health-related feedback to the user. The LCD output display stage also helps in verifying whether the monitoring system is functioning correctly. It acts as a quick local monitoring interface and supports the portability of the system in home healthcare applications.

Fig 4.7 LCD Output Display

The above figure 4.7 shows the LCD output display of the proposed monitoring system. It presents the real-time heart rate and SpO_2 values generated after the data processing stage.

4.3.4 Cloud Upload Process

The final stage of the software pipeline is the cloud upload process, where the processed physiological data is transmitted from the ESP32 to the Thing Speak cloud platform using WiFi communication.

The embedded program first checks whether the ESP32 is connected to a valid WiFi network. Once the connection is confirmed, the software constructs an HTTP request containing the API key and the monitored values. This request is then sent to the Thing Speak server. The cloud upload process allows the system to store the monitored values online and generate graphical visualizations for remote observation. It also supports long-term data tracking and record maintenance. This stage is crucial because it transforms the monitoring system from a simple local device into an IoT-enabled healthcare platform, making the project more advanced and practical.

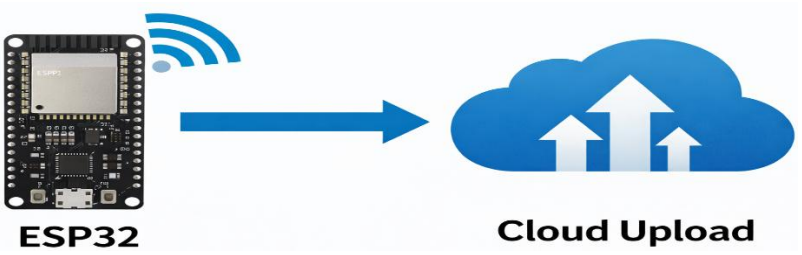

Fig 4.8 ESP32 → Cloud upload diagram

The above figure 4.8 shows the cloud upload process used in the proposed system. It illustrates how the processed physiological data is transmitted from the ESP32 to the Thing Speak cloud platform for storage and visualization.

4.4 Real-Time Data Acquisition Module

The Real-Time Data Acquisition Module is one of the most important parts of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare. This module is responsible for continuously collecting physiological signals from the user and converting them into meaningful health-related data for monitoring and analysis. In the proposed system, real-time data acquisition is achieved using the IR sensor, which senses physiological signals from the user's body.

These signals are captured continuously and sent to the ESP32 microcontroller for processing. Since sleep apnea monitoring requires continuous observation of body-related parameters during sleep, the real-time data acquisition module plays a critical role in ensuring uninterrupted monitoring.

The module is designed to operate continuously so that the monitored values can be updated at regular intervals without manual intervention. This helps in detecting variations in physiological parameters over time and provides a more realistic view of the user's health condition during rest or sleep. The acquired sensor data is used for calculating important physiological parameters such as heart rate and SpO_2, which are relevant for observing sleep-related abnormalities.

Once the data is collected and processed, it is displayed locally on the LCD display and transmitted to the Thing Speak cloud platform for remote access and graphical visualization. The main advantage of this module is that it allows the system to perform continuous and automatic health monitoring in a home environment. This makes the proposed system more practical, portable, and suitable for real-time healthcare applications. Thus, the real-time data acquisition module forms the sensing foundation of the proposed monitoring system and enables continuous collection of physiological data for analysis and observation.

4.4.1 Heat Rate and Spo2 Monitoring

The Heart Rate Monitoring and SpO_2 Monitoring section represents the physiological monitoring function of the proposed system. These two parameters are important indicators of a person's respiratory and cardiovascular condition, especially in the context of sleep apnea monitoring. The heart rate refers to the number of heart beats per minute, while SpO_2 (oxygen saturation) represents the percentage of oxygen carried in the blood. During sleep apnea episodes, irregular breathing patterns may affect oxygen levels and heart rhythm, making these parameters useful for observation and early detection of abnormal conditions. In the proposed system, the IR sensor is used to collect physiological signals from the user's body.

These signals are read and processed by the ESP32 microcontroller, which estimates the heart rate and SpO_2-related values in real time. The monitored values are then displayed on the LCD display and also uploaded to the Thing Speak cloud platform for remote access and graph generation. The continuous monitoring of heart rate and SpO_2 helps in understanding the user's physiological state during operation. If the values show unusual variation, the user or caregiver can identify that the physiological condition requires further observation. This monitoring mechanism is useful because it provides immediate local output and long-term cloud-based tracking of important health-related parameters. It supports the objective of the proposed project by enabling real-time, portable, and IoT-based observation of sleep apnea-related physiological changes.

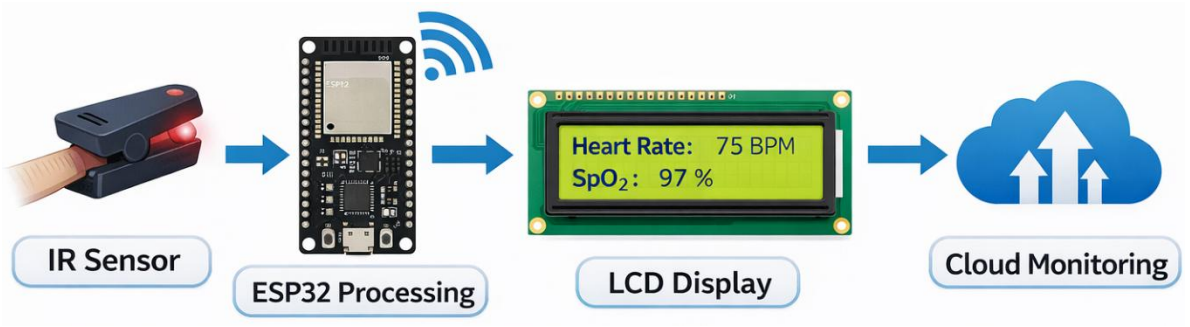

Fig 4.9 Heart Rate and SpO_2 Monitoring Process

The above figure 4.9 illustrates the heart rate and SpO_2 monitoring process of the proposed system. It shows how physiological signals are sensed, processed by the ESP32, displayed locally, and transmitted to the cloud platform for real-time health monitoring.

4.5 Physiological Parameter Monitoring

The Physiological Parameter Monitoring section of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare focuses on the observation and interpretation of the key health-related values measured by the system. In this project, the main physiological parameters considered are heart rate and SpO_2 (oxygen saturation), as these values are important indicators of respiratory and cardiovascular condition. Sleep apnea is often associated with irregular breathing patterns during sleep, which may influence oxygen saturation and heart activity. Therefore, continuous monitoring of these physiological parameters is useful for identifying abnormal variations and understanding the user's health condition during rest. The proposed system continuously acquires sensor signals using the IR sensor and processes them using the ESP32 microcontroller.

The resulting heart rate and SpO_2 values are then displayed on the LCD display and uploaded to the Thing Speak cloud platform for real-time and long-term observation. Monitoring physiological parameters in real time provides several benefits. It allows the user to immediately observe health-related changes, supports cloud-based trend analysis, and improves the practicality of home healthcare monitoring.

4.5.1 Heart Rate & Spo2 Parameters

The two major physiological parameters monitored in the proposed system are heart rate and SpO_2 (oxygen saturation). These parameters are selected because they provide useful information related to the user's breathing and circulatory condition, which are relevant in the context of sleep apnea observation.

Heart Rate Parameter

The heart rate represents the number of times the heart beats per minute (BPM – Beats Per Minute). It is one of the basic physiological indicators used in healthcare monitoring. A normal resting heart rate generally falls within a moderate range, but unusual variations may indicate physiological stress, irregular activity, or changes associated with disturbed sleep patterns. In the proposed system, heart rate is estimated from the physiological signal captured by the IR sensor. The signal is processed by the ESP32, which detects pulse-related changes and calculates the heart rate value. This value is then shown on the LCD display and also uploaded to the cloud platform. Continuous heart rate observation improves the effectiveness of real-time health monitoring. It also provides useful information about the cardiovascular response of the patient during sleep. Monitoring heart rate regularly helps in detecting possible irregularities at an early stage.

SpO_2 Parameter

The SpO_2 value represents the percentage of oxygen present in the blood. It is an important health parameter because it reflects how effectively oxygen is being carried to body tissues. In sleep apnea-related conditions, oxygen levels may fluctuate due to interrupted breathing, making SpO_2 monitoring highly relevant. In the proposed system, SpO_2-related signal variation is observed through the physiological sensing mechanism and processed by the microcontroller. The estimated SpO_2 value is displayed locally and stored in the cloud for further observation and graph generation. A drop in oxygen level during sleep may indicate abnormal respiratory activity. Continuous SpO_2 observation helps in identifying possible apnea-related conditions.

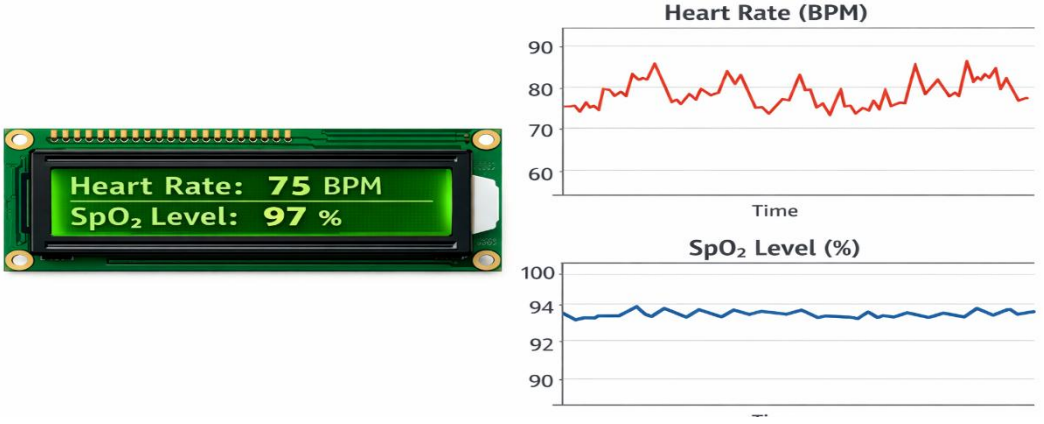

Fig 4.10 Heart Rate and SpO_2 Parameters

The above figure 4.10 represents the physiological parameters monitored in the proposed system.

4.5.2 Health Data Interpretation

The Health Data Interpretation stage involves understanding and analyzing the monitored physiological values generated by the proposed system. After the heart rate and SpO_2 values are collected, processed, displayed, and stored in the cloud, the next important step is to interpret what these values indicate about the user's physiological condition.

The proposed system allows health data interpretation in two major ways:

1. Local interpretation using the LCD display.
2. Remote interpretation using the Thing Speak cloud graphs and stored records.

The LCD display provides immediate access to the current physiological readings, allowing the user to directly observe the values in real time. This helps in quick understanding of whether the monitored values appear stable or show sudden variation. The Thing Speak cloud platform provides a more detailed interpretation by displaying the monitored values in the form of graphs and time-based data records. By observing the graph patterns, the user can identify whether the heart rate or SpO_2 values remain consistent or fluctuate over time. This interpretation is useful in-home healthcare monitoring because it helps the user or caregiver observe trends rather than relying only on a single reading. It also supports better awareness of the physiological condition during monitoring periods. Although the proposed system is designed mainly for educational and monitoring purposes rather than clinical diagnosis, the interpretation of collected data helps demonstrate how IoT-based health monitoring can support real-time observation and early awareness of physiological variations.

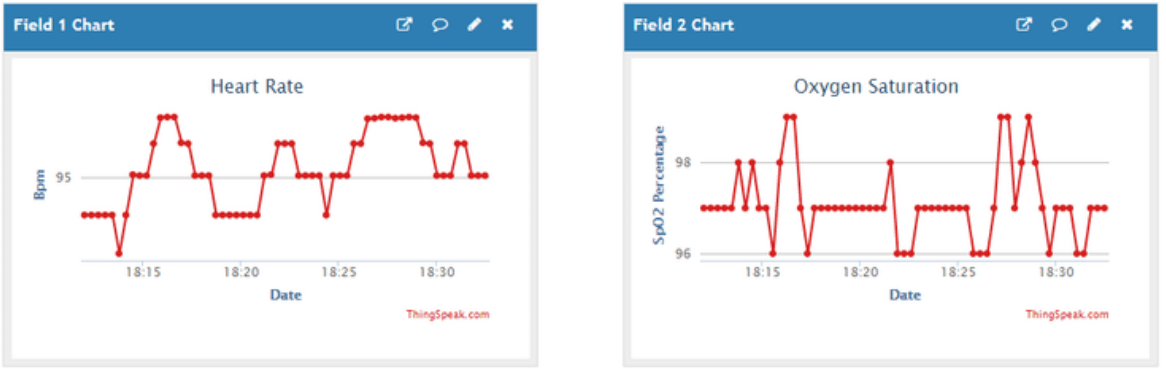

Fig 4.11 Health Data Interpretation

The above figure 4.11 illustrates the health data interpretation process of the proposed system. It shows how heart rate and SpO_2 values can be analyzed through graphical visualization and time-based monitoring for better physiological understanding.

CHAPTER 5

RESULT AND DISCUSSION

5.1 Experimental Setup

The experimental setup of the proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare was developed to evaluate the performance of the hardware and software modules under real-time operating conditions. The setup was designed to demonstrate how physiological parameters such as heart rate and SpO_2 can be sensed, processed, displayed, and transmitted using the proposed IoT-based architecture. The experimental arrangement includes the IR sensor, ESP32 microcontroller, LCD display, power supply unit, and WiFi-based cloud connectivity. These components were connected and configured to perform continuous monitoring of physiological signals from the user. The sensor captured the input signal, which was then processed by the ESP32 and displayed on the LCD screen. At the same time, the ESP32 transmitted the monitored values to the Thing Speak cloud platform using WiFi communication. The uploaded data was stored in the cloud and visualized in the form of graphs, making it possible to analyse the trend of the monitored physiological parameters over time. The experimental setup was tested under controlled demonstration conditions to verify whether the system was capable.

- Capturing physiological input,
- Generating heart rate and SpO_2 values,
- Displaying real-time output,
- Transmitting data to the cloud,
- And maintaining a continuous monitoring flow.

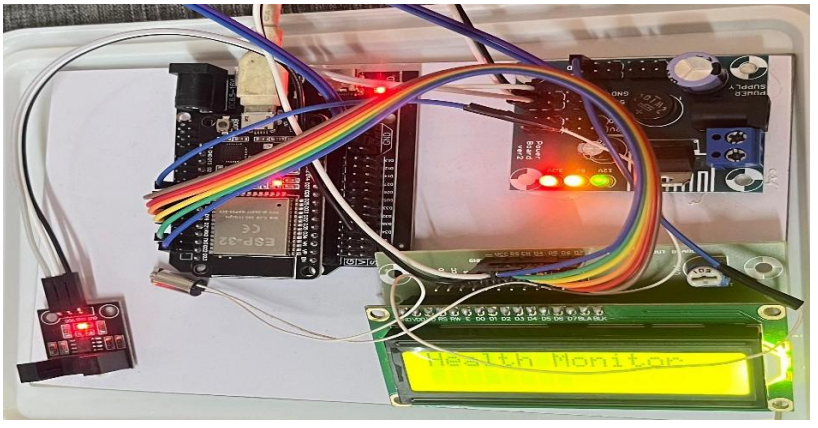

Fig 5.1 Experimental Setup

The above figure 5.1 shows the experimental setup of the proposed monitoring system. It includes the ESP32 microcontroller, IR sensor, LCD display, and supporting components used for real-time physiological monitoring and cloud-based data transmission.

5.2 Output

Fig 5.2 LCD Output Result

The above figure5.2 shows the LCD output of the proposed system. It displays the real-time heart rate and SpO_2 values generated during physiological monitoring.

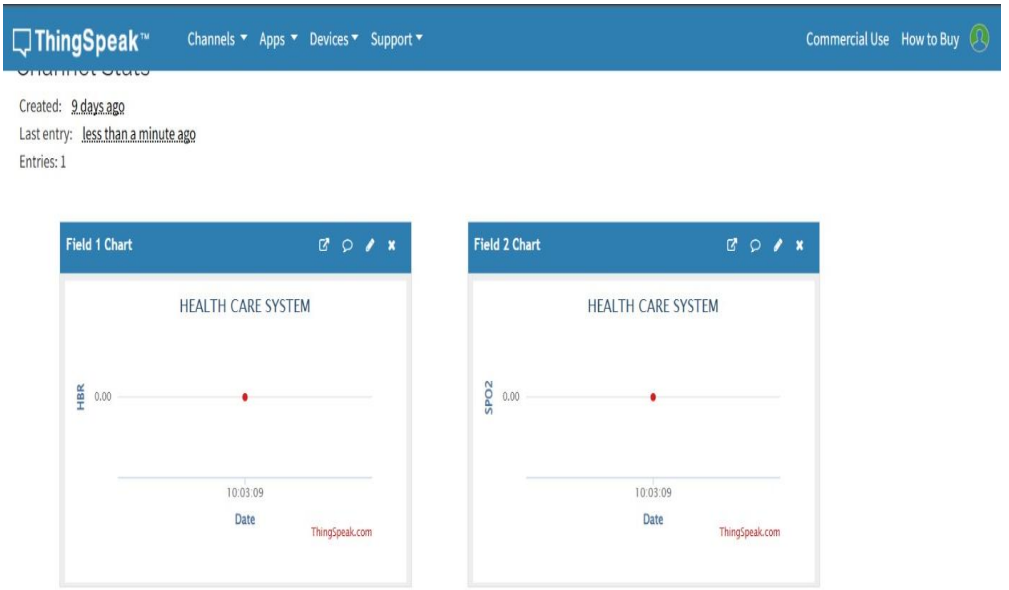

Fig 5.3 Thing Speak Output Result

The above figure 5.3 shows the Thing Speak cloud output of the proposed system. It displays the uploaded physiological values in graphical and field-based format for remote monitoring.

Fig 5.4 Heart Rate Graph

The above figure5.4 illustrates the heart rate graph generated by the proposed system. It represents the variation of heart rate values over time during monitoring.

Fig 5.5 SpO_2 Graph

The above figure 5.5 illustrates the SpO_2 graph generated by the proposed system. It represents the variation of oxygen saturation values during the monitoring process.

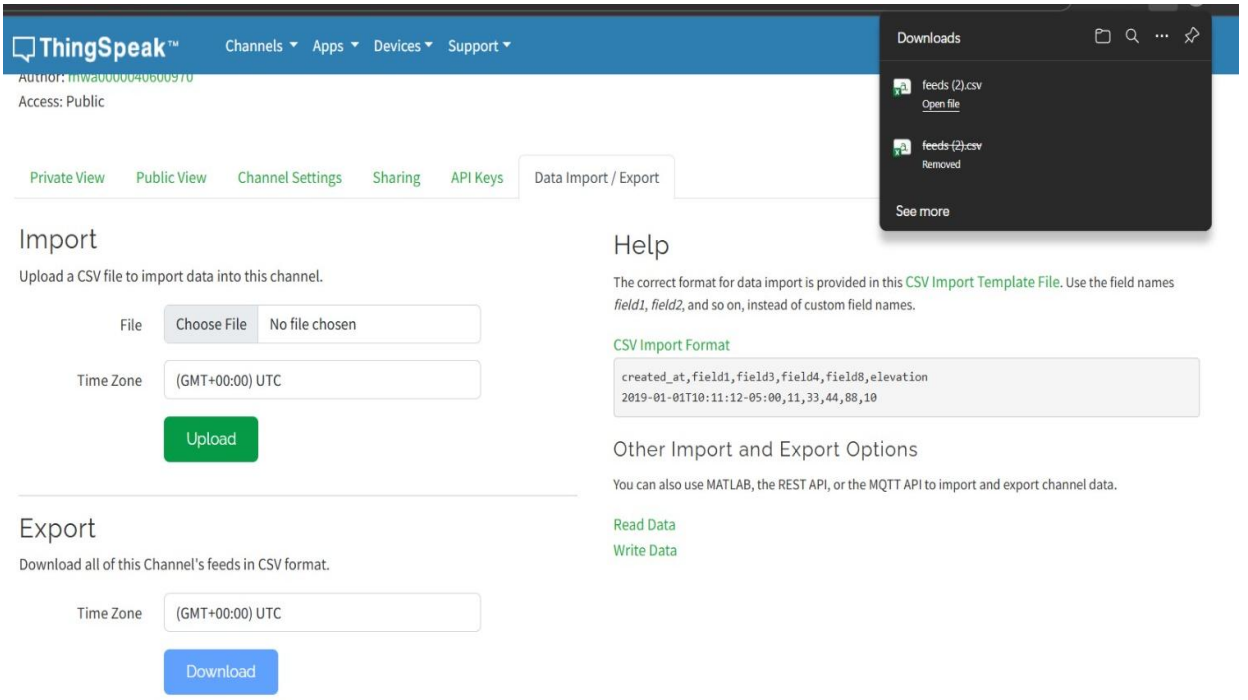

Fig 5.6 Excel Data Download

The above figure 5.6 shows the downloadable Excel data generated by the proposed system. It contains the monitored physiological values stored for future reference and analysis.

5.3 Achievements of The Project

The proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare successfully achieved the primary objectives defined during the design and implementation stages. The project was developed to create a portable, cost-effective, and IoT-enabled physiological monitoring system capable of observing health-related parameters in a home environment. One of the major achievements of the project is the successful integration of hardware components such as the IR sensor, ESP32 microcontroller, LCD display, and WiFi communication into a single functional monitoring system. The system was able to continuously collect physiological signals and generate useful output values. Another important achievement is the implementation of real-time local display output, which allows the user to directly observe heart rate and SpO_2 values on the LCD screen. This improves the usability of the system and provides immediate health-related feedback. The project also successfully implemented cloud-based monitoring using the Thing Speak platform, which enabled remote visualization and storage of physiological data. The ability to generate graphs and download Excel records further improved the practicality and analytical value of the system.

In addition, the project demonstrates the applicability of IoT technology in healthcare monitoring, particularly for educational and home-based observation purposes. The overall system was able to perform sensing, processing, display, and data transmission in an organized and reliable manner. Thus, the proposed project achieved its intended objectives and provides a meaningful demonstration of a portable IoT-based healthcare monitoring system.

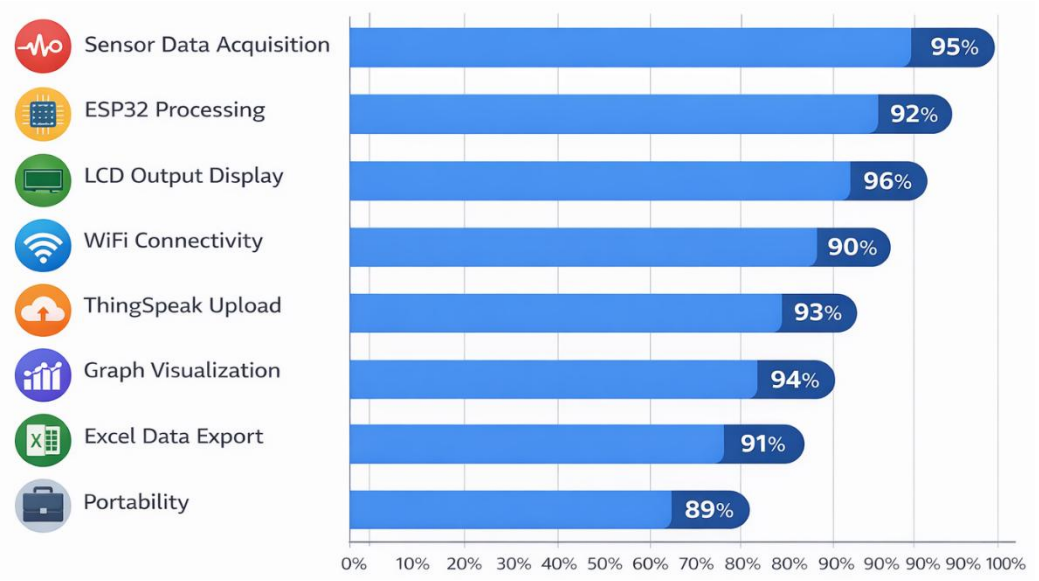

Fig 5.7 Final System Output

The above figure 5.7 represents the final output of the proposed monitoring system. It combines the local display, cloud output, and graphical results to demonstrate the successful operation of the project.

CHAPTER 6

CONCLUSION

The proposed Portable IoT-Based Sleep Apnea Monitoring System for Home Healthcare was successfully designed and implemented to demonstrate the concept of real-time physiological monitoring using IoT technology. The project was developed with the objective of monitoring important health-related parameters such as heart rate and SpO_2 in a portable and user-friendly manner suitable for home healthcare applications.

The system integrates essential hardware components such as the IR sensor, ESP32 microcontroller, LCD display, power supply unit, and WiFi communication, along with software tools such as the Arduino IDE and the Thing Speak cloud platform. Through this integration, the system was able to collect physiological signals, process the values, display real-time output locally, and transmit the monitored data to the cloud for remote observation and graphical analysis.

The implementation and results of the project showed that the proposed system can effectively perform real-time monitoring of physiological parameters and support cloud-based data visualization and storage. The generated outputs, including LCD display readings, Thing Speak graphs, and Excel data download, demonstrate the successful working of the project and its practical relevance in the field of IoT-based healthcare monitoring.

One of the major advantages of the proposed system is its portability, low cost, and ease of use, which make it suitable for educational demonstration and home monitoring applications. The project also highlights how embedded systems and cloud platforms can be combined to create smart and accessible healthcare solutions.

Although the proposed system is mainly intended for educational and prototype-level implementation, it provides a strong foundation for future enhancement in the field of healthcare monitoring. With further improvements such as advanced sensors, alert systems, mobile application integration, and data analytics, the system can be extended into a more intelligent and comprehensive health monitoring solution. Thus, the proposed project successfully achieved its objectives and demonstrates the practical application of IoT technology in portable physiological monitoring for home healthcare.

APPENDIX

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SOURCE CODE

WiFi Initialization Code

```
#include <WiFi.h>
#include <HTTPClient.h>

const char* ssid = "YOUR_SSID";
const char* password = "YOUR_PASSWORD";

const char* server = "http://api.thingspeak.com/update";
String apiKey = "YOUR_THINGSPEAK_WRITE_API_KEY";
```

IR Sensor Reading Code

```
const int IR_PIN = 34;

void setup() {
  Serial.begin(115200);
  pinMode(IR_PIN, INPUT);
}
```

Heart Rate Calculation Code

```
unsigned long lastBeat = 0;
int beatCount = 0;
float heartRate = 0;

void loop() {
  int sensorValue = analogRead(IR_PIN);

  if (sensorValue > 2000) {
    if (millis() - lastBeat > 300) {
      beatCount++;
      lastBeat = millis();
    }
  }
}
```

ThingSpeak Cloud Upload Code

```
if (WiFi.status() == WL_CONNECTED) {
  HTTPClient http;
  String url = String(server) + "?api_key=" + apiKey + "&field1=" + heartRate;
  http.begin(url);
  int httpResponseCode = http.GET();
```

```
  http.end();
}


 #include <WiFi.h>
#include <HTTPClient.h>

// Replace with your WiFi credentials
const char* ssid = "YOUR_SSID";
const char* password = "YOUR_PASSWORD";

// ThingSpeak settings
const char* server = "http://api.thingspeak.com/update";
String apiKey = "YOUR_THINGSPEAK_WRITE_API_KEY";

// IR sensor pin
const int IR_PIN = 34;

// Variables for heart rate calculation
unsigned long lastBeat = 0;
int beatCount = 0;
float heartRate = 0;

void setup() {
  Serial.begin(115200);
  pinMode(IR_PIN, INPUT);

  // Connect to WiFi
  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");

  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");
  }

  Serial.println("\nConnected to WiFi");
}

void loop() {
  int sensorValue = analogRead(IR_PIN); // Read IR sensor value

  // Simple threshold-based heartbeat detection
  if (sensorValue > 2000) {   // Adjust threshold based on your sensor
    if (millis() - lastBeat > 300) { // Avoid multiple triggers per beat
      beatCount++;
      lastBeat = millis();
    }
  }

  // Calculate heart rate every 15 seconds
  static unsigned long lastTime = millis();
```

```
  if (millis() - lastTime >= 15000) {
    heartRate = (beatCount * 4); // Approximate BPM
    beatCount = 0;
    lastTime = millis();

    Serial.print("Heart Rate: ");
    Serial.println(heartRate);

    // Send data to ThingSpeak
    if (WiFi.status() == WL_CONNECTED) {
      HTTPClient http;

      String url = String(server) + "?api_key=" + apiKey + "&field1=" + heartRate;
      http.begin(url);

      int httpResponseCode = http.GET();

      if (httpResponseCode > 0) {
        Serial.println("Data sent to ThingSpeak successfully");
      } else {
        Serial.println("Error sending data to ThingSpeak");
      }

      http.end();
    } else {
      Serial.println("WiFi Disconnected");
    }
  }

  delay(10);
}
```